

NIPTELL

Classifier-Free Diffusion Guidance (CFG)

Contents

- ✓ Motivation for Classifier free Diffusion Guidance (CFG)
- ✓ Concept of CFG
- ✓ Joint Training of Neural Network
- ✓ Mathematical Formulation of CFG
- ✓ Sampling Process
- ✓ Comparison between Classifier Guided diffusion and Classifier free diffusion guidance

Motivation for Classifier Free Diffusion Guidance

Requirement for a Separate Model:

- It requires training an additional classifier specifically on noisy images x_t to provide the necessary gradients at every step of the diffusion process
- Standard off the shelf classifiers like Resnet trained on ImageNet will fail as they are trained on clear images
- Complex pipeline as two separate networks (Unet/Diffusion network and the classifier) needs to be trained.
- Classifier gradients must be computed at every timestep, increasing computational cost during sampling

Concept of Classifier-Free Diffusion Guidance (CFG)

Introduced in the paper “Classifier-Free Diffusion Guidance“ by Jonathan Ho and Tim Salimans, from Google Research, Brain Team

Core Idea

- Conditional generation without training an external classifier.
- In classifier-free guidance, a single diffusion model is trained to perform both conditional and unconditional generation simultaneously.
- The diffusion model learns both conditional and unconditional denoising, allowing guidance to be obtained without a separate classifier.

Ref:

Ho et al., Classifier-Free Diffusion Guidance NeurIPS Workshop (2022)

Joint Training of Neural network

Class labels are
supplied
for conditional
training

Neural Network

Conditional score

Unconditional score

Class labels are dropped with a
fixed probability (usually 10 to
20 percent)
Class labels are replaced by null
token

Train a conditioned diffusion model $\epsilon_{\theta}(x_t, t, y)$ with condition dropout $\epsilon_{\theta}(x_t, t, \phi)$

y is removed to make
the model unconditional

ϕ = Null token
 ϕ represents absence of
conditioning information

Mathematical Formulation

$$p(y | x_t)$$



The probability of getting class label y given the noisy image x_t

Using Bayes' theorem:

$$p(y | x_t) = \frac{p(x_t | y)p(y)}{p(x_t)}$$

Taking logarithm:

$$\log p(y | x_t) = \log p(x_t | y) + \log p(y) - \log p(x_t)$$

- ✓ Taking gradient with respect to x_t on both sides.
- ✓ Since $p(y)$ is constant (doesn't depend on x_t), its gradient w.r.t x_t is zero.

$$\nabla_{x_t} \log p(y | x_t) = \nabla_{x_t} \log p(x_t | y) + \nabla_{x_t} \log p(x_t)$$

Conditional term

Unconditional term

CFG approximates this classifier gradient using the diffusion model itself.

Noise Prediction

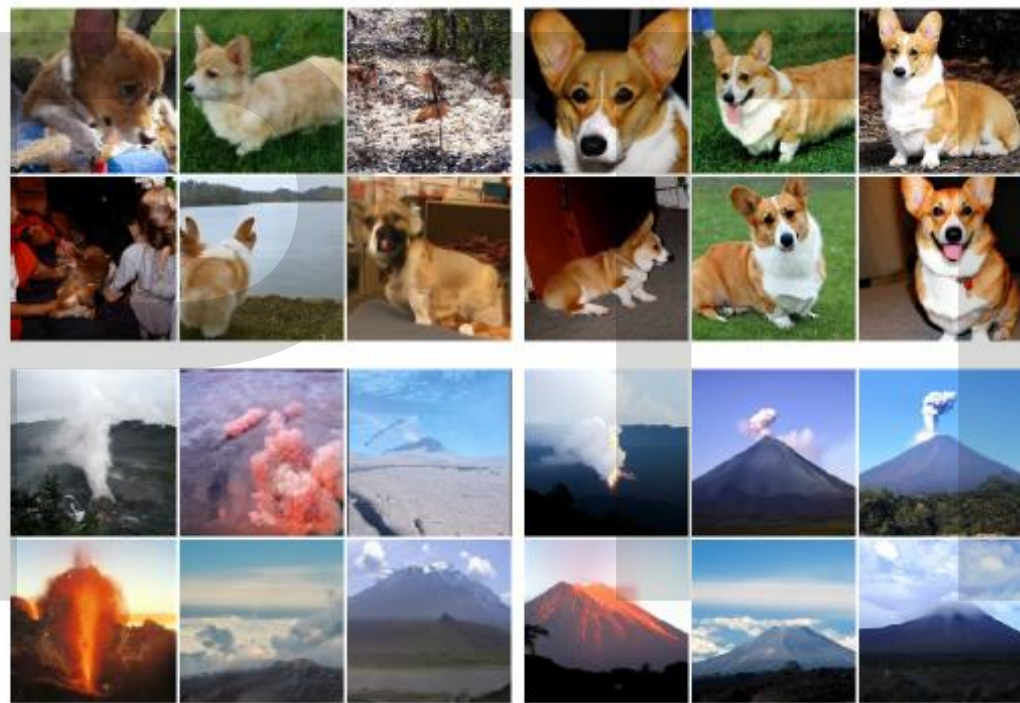
$$\hat{\epsilon}(x_t, t, y) = (1 + s)\epsilon_\theta(x_t, t, y) - s\epsilon_\theta(x_t, t, \emptyset)$$

s = classifier guidance

Sampling from classifier free guidance model

- ✓ Start by sampling data with conditioning from the dataset
- ✓ Randomly discard the conditioning to train the unconditional version of the same network
- ✓ Train the same network using both conditional and unconditional examples.
- ✓ This teaches the diffusion model to be able to guide the model towards generating samples from a class without explicitly needing a classifier

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Non Guided Sample

Classifier free guided
samples with $s=3.0$

Ref:
Ho et al., Classifier-Free Diffusion Guidance NeurIPS Workshop (2022)

Comparison between Classifier Guided and Classifier free Guidance

Feature	Classifier Guided	Classifier free Guidance
Separate Classifier	YES	NO
Noisy Classifier training	Required	Not required
Guidance score	Classifier gradient	Conditional –Unconditional difference
Sampling Cost	U-Net+ Classifier	Two U-Net Passes

NPTEL

Summary

NPTTEL

Next Session
Stable Diffusion Model